EXPERIMENT 10 - PYTHON CODE

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# EXPERIMENT 10 - Resolution Refutation
clauses = [
    {"EnjoysLearning"},
    {"HasFreeTime"},
    {"NotEnjoysLearning", "Happy"},
    {"NotHasFreeTime", "Happy"}
]
goal = "Happy"
clauses.append({"Not" + goal})

def complement(literal):
    return literal[3:] if literal.startswith("Not") else "Not" + literal

def display(clause):
    return " OR ".join(sorted(clause)) if clause else "EMPTY"

print("RESOLUTION REFUTATION")
print("=" * 35)
print("\nStep 1: Knowledge Base in CNF")
for i, clause in enumerate(clauses[:-1], 1):
    print(f"  C{i}: {display(clause)}")
print("\nStep 2: Negate the Goal")
print("  Goal:", goal)
print("  Added clause: NotHappy")

print("\nStep 3: Apply Resolution")
known = [frozenset(c) for c in clauses]
seen = set(known)
empty_found = False
step = 1
while True:
    new_clauses = []
    for i in range(len(known)):
        for j in range(i + 1, len(known)):
            left, right = set(known[i]), set(known[j])
            for literal in sorted(left):
                opposite = complement(literal)
                if opposite in right:
                    result = (left - {literal}) | (right - {opposite})
                    print(f"  Step {step}: Resolve [{display(left)}] and [{display(right)}] on {literal}")
                    print("          Result:", display(result))
                    step += 1
                    if not result:
                        empty_found = True
                        break
                    frozen = frozenset(result)
                    if frozen not in seen:
                        seen.add(frozen)
                        new_clauses.append(frozen)
                if empty_found:
                    break
            if empty_found:
                break
        if empty_found:
            break
    if empty_found or not new_clauses:
        break
    known.extend(new_clauses)

print("\n--- Final Result ---")
if empty_found:
    print("Empty clause derived: Yes")
    print("Contradiction found.")
    print("Therefore, the goal 'Happy' is proved.")
else:
    print("Empty clause derived: No")
    print("The goal could not be proved by this resolution run.")
```